

Hospital Patient & Billing Analytics Dashboard

Operational, Clinical & Financial Performance Tracking Specification

1. PROJECT TITLE

Hospital Patient & Billing Analytics Dashboard

2. PROJECT OBJECTIVE

To analyze patient admission, treatment, and billing data in order to build an interactive Power BI dashboard that helps hospital management monitor patient trends, admission patterns, and revenue performance to support operational and financial decision-making.

3. BUSINESS PROBLEM / OPPORTUNITY

Hospital administrators currently lack a unified view of patient demographics, admission types, medical conditions, and billing outcomes. Without this visibility, it is difficult to identify capacity bottlenecks, high-cost conditions, insurance trends, or opportunities to improve patient care efficiency.

4. TARGET AUDIENCE

Hospital administrators, department heads, and healthcare analysts who need to monitor patient volumes, billing performance, and care outcomes.

5. SCOPE

In scope:

- Patient demographics, admission types, medical conditions
- Length of stay, billing amounts, insurance providers
- Hospital and doctor metadata

Out of scope:

- Real-time clinical data and treatment protocols
- Individual patient outcomes tracking
- External claims data

6. DATA SOURCE(S)

Kaggle "Healthcare Dataset" (synthetic) - patient-level records including Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, Insurance Provider, Billing Amount, Room Number, Admission Type, Discharge Date, Medication, and Test Results. Provided in CSV format.

7. KEY METRICS / KPIS

- Total Patients Admitted
- Average Billing Amount / Total Revenue
- Average Length of Stay

- Admissions by Medical Condition
- Admissions by Type (Emergency, Elective, Urgent)
- Insurance Provider Distribution
- Test Result Outcomes (Normal / Abnormal / Inconclusive)

8. DELIVERABLES

- Business Requirements Document (BRD)
- Functional Requirements Document (FRD)
- Dataset Assessment
- Cleaned Dataset
- Dashboard Mockups
- Power BI Dashboard (.pbix)
- Data Model Documentation
- Analysis Report
- README

9. TIMELINE / MILESTONES

Day	Milestone
Day 1	Business Requirements Document (BRD)
Day 2	Dataset Assessment & Cleaning
Day 3	Functional Requirements Document (FRD) & Dashboard Mockups
Day 4	Power BI Dashboard Build (Draft)
Day 5	Finalize Dashboard, Data Model & Analysis Report

10. NOTES / ASSUMPTIONS

- Dataset is synthetic; contains no real patient personally identifiable information (PII)
- Billing amounts are assumed to be in USD
- “Date of Admission” and “Discharge Date” will be used to calculate Length of Stay
- No external claims or clinical outcome data will be joined to this dataset